

PREFACE FOR FACULTY

The Z-Transform extends continuous-frequency Fourier analysis into the complex plane $z = re^{j\omega}$, enabling rigorous analysis of unstable systems, transient dynamics, difference equations, and system stability.

Pedagogical Strategy:

1. Define the two-sided (bilateral) Z-transform: $X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n}$.
 2. Show that the DTFT is the special case of the Z-transform evaluated on the unit circle ($r = 1 \implies z = e^{j\omega}$).
 3. Emphasize that **a Z-transform is incomplete without specifying its Region of Convergence (ROC)**.
 4. Establish the 6 fundamental properties of the ROC (connected ring, no poles inside ROC, right-sided $\implies |z| > r_{\max}$, left-sided $\implies |z| < r_{\min}$, two-sided $\implies r_1 < |z| < r_2$).
 5. Derive standard transform pairs and properties (Linearity, Shifting, Scaling, Differentiation in z-domain, Convolution).
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1. LEARNING OBJECTIVES

By the end of this lecture, students will be able to:

1. **Derive** the Z-transform and sketch the ROC for causal, anti-causal, and two-sided sequences.
 2. **Apply** the fundamental properties of the ROC to deduce stability and causality.
 3. **Utilize** Z-transform algebraic properties to transform composite discrete signals.
 4. **Relate** pole-zero configurations in the complex z -plane to time-domain signal behaviors.
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2. MATHEMATICAL FOUNDATIONS

2.1 The Bilateral Z-Transform

$$X(z) = \mathcal{Z}x[n] = \sum_{n=-\infty}^{\infty} x[n]z^{-n}, \quad z = re^{j\omega} \in \mathbb{C}$$

Substituting $z = re^{j\omega}$:

$$X(re^{j\omega}) = \sum_{n=-\infty}^{\infty} (x[n]r^{-n})e^{-j\omega n} = \mathcal{F}x[n]r^{-n}$$

The Z-transform is the DTFT of the sequence multiplied by a real exponential damping factor r^{-n} .

2.2 Region of Convergence (ROC) Properties

The ROC is the set of points in the z -plane for which $\sum_{n=-\infty}^{\infty} |x[n]r^{-n}| < \infty$:

1. The ROC is a ring or disk centered at the origin: $r_R < |z| < r_L$.
2. The ROC **cannot contain any poles**.
3. For a finite-duration sequence, the ROC is the entire z -plane, except possibly $z = 0$ and/or $z = \infty$.
4. For a **right-sided (causal)** sequence, the ROC extends outward from the outermost finite pole: $|z| > r_{\max}$.
5. For a **left-sided (anti-causal)** sequence, the ROC extends inward from the innermost non-zero pole: $|z| < r_{\min}$.
6. For a **two-sided** sequence, the ROC is an open annular ring bounded by poles: $r_1 < |z| < r_2$.
7. An LTI system is **BIBO stable** if and only if the ROC of its system function $H(z)$ **includes the unit circle** $|z| = 1$.

2.3 Common Z-Transform Pairs

Sequence $x[n]$	Z-Transform $X(z)$	ROC
$\delta[n]$	1	Entire z -plane
$\delta[n - n_0]$	z^{-n_0}	All z , except $z = 0$ ($n_0 > 0$) or $z = \infty$ ($n_0 < 0$)
$u[n]$	$\frac{1}{1-z^{-1}} = \frac{z}{z-1}$	\$
$-u[-n - 1]$	$\frac{1}{1-z^{-1}} = \frac{z}{z-1}$	\$
$a^n u[n]$	$\frac{1}{1-az^{-1}} = \frac{z}{z-a}$	\$
$-a^n u[-n - 1]$	$\frac{1}{1-az^{-1}} = \frac{z}{z-a}$	\$
$na^n u[n]$	$\frac{az^{-1}}{(1-az^{-1})^2} = \frac{az}{(z-a)^2}$	\$
$\cos(\omega_0 n) u[n]$	$\frac{1 - \cos \omega_0 z^{-1}}{1 - 2 \cos \omega_0 z^{-1} + z^{-2}}$	\$
$\sin(\omega_0 n) u[n]$	$\frac{\sin \omega_0 z^{-1}}{1 - 2 \cos \omega_0 z^{-1} + z^{-2}}$	\$

3. WORKED NUMERICAL EXAMPLES

Example 5.1: Two-Sided Sequence Z-Transform

Problem: Find the Z-transform and ROC of $x[n] = \left(\frac{1}{2}\right)^n u[n] + \left(-\frac{1}{3}\right)^n u[-n - 1]$.

Solution: Let $x_1[n] = (1/2)^n u[n]$ and $x_2[n] = -[-(-1/3)^n u[-n - 1]]$:

- $X_1(z) = \frac{1}{1 - \frac{1}{2}z^{-1}}$ with ROC: $|z| > \frac{1}{2}$.
- $X_2(z) = \frac{1}{1 - (-\frac{1}{3})z^{-1}} = \frac{1}{1 + \frac{1}{3}z^{-1}}$ with ROC: $|z| < |-\frac{1}{3}| = \frac{1}{3}$. The overall ROC is the intersection: $\text{ROC} = |z| > 1/2 \cap |z| < 1/3 = \emptyset$ (Empty Set). Because the two regions do not overlap, the Z-transform $X(z)$ **does not exist**.

Example 5.2: Two-Sided Sequence with Valid Annular ROC

Problem: Find $X(z)$ and sketch ROC for $x[n] = (\frac{1}{3})^n u[n] - 2^n u[-n - 1]$.

Solution:

- $X_1(z) = \frac{1}{1 - \frac{1}{3}z^{-1}}$ with $\text{ROC}_1 : |z| > 1/3$.
- $X_2(z) = \frac{1}{1 - 2z^{-1}}$ with $\text{ROC}_2 : |z| < 2$. Overall ROC: $\text{ROC}_1 \cap \text{ROC}_2 = \frac{1}{3} < |z| < 2$.

$$X(z) = \frac{1}{1 - \frac{1}{3}z^{-1}} + \frac{1}{1 - 2z^{-1}} = \frac{(1 - 2z^{-1}) + (1 - \frac{1}{3}z^{-1})}{(1 - \frac{1}{3}z^{-1})(1 - 2z^{-1})} = \frac{2 - \frac{7}{3}z^{-1}}{(1 - \frac{1}{3}z^{-1})(1 - 2z^{-1})}$$

Poles at $z = 1/3$ and $z = 2$. Zeros at $z = 7/6 \approx 1.167$. Since the annular ring $\frac{1}{3} < |z| < 2$ includes the unit circle $|z| = 1$, the signal possesses a valid DTFT and is stable.

4. UNIVERSITY EXAMINATION QUESTIONS & MARKING RUBRIC

Question 1 (15 Marks)

(a) List the properties of the Region of Convergence (ROC) of the Z-transform. (6 Marks) **(b)** Find the Z-transform and sketch the pole-zero plot and ROC of:

$$x[n] = (0.5)^n u[n] + (0.8)^n u[n]$$

Is the system stable? Is it causal? Justify. (9 Marks)

Model Answer & Step-by-Step Marking Rubric:

- **Part (a):**
 - Listing 6 properties with brief physical justifications (6 Marks)
- **Part (b):**
 - Transform of individual terms:

$$X(z) = \frac{1}{1 - 0.5z^{-1}} + \frac{1}{1 - 0.8z^{-1}}, \quad \text{ROC}_1 : |z| > 0.5, ; \text{ROC}_2 : |z| > 0.8$$

(3 Marks)

- Combine into rational fraction:

$$X(z) = \frac{(1 - 0.8z^{-1}) + (1 - 0.5z^{-1})}{(1 - 0.5z^{-1})(1 - 0.8z^{-1})} = \frac{2 - 1.3z^{-1}}{1 - 1.3z^{-1} + 0.4z^{-2}} = \frac{2z(z - 0.65)}{(z - 0.5)(z - 0.8)}$$

(2 Marks)

- Poles: $z = 0.5, 0.8$. Zeros: $z = 0, 0.65$.
- Overall ROC: $|z| > \max(0.5, 0.8) = |z| > 0.8$. (2 Marks)

- Causality & Stability:
 - **Causal:** Yes, because the ROC is outside the outermost pole $|z| > 0.8$.
 - **Stable:** Yes, because the ROC $|z| > 0.8$ includes the unit circle $|z| = 1$. (2 Marks)
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5. PYTHON VERIFICATION SCRIPT

```
import numpy as np
import scipy.signal as signal

# Pole-zero calculation
b = [2, -1.3]
a = [1, -1.3, 0.4]
zeros, poles, gain = signal.tf2zpk(b, a)
print(f"Poles: {poles}")
print(f"Zeros: {zeros}")
```